

EDUCATION

- **The University of Chicago, Booth School of Business** Chicago, IL
Ph.D. student in Management Science / Operations Management Sep.2025 –
- **The University of Chicago** Chicago, IL
Master of Science in Computational and Applied Math Sep.2024 – Jun.2025
- **Hong Kong University of Science and Technology** Hong Kong SAR
Bachelor of Science in (Pure) Mathematics (First Class Honors) Sep.2020 – Jun.2024
- **Columbia University in the City of New York** New York City, NY
Exchange Student at Columbia College Jan.2023 – May.2023

PUBLICATION

- Haoyu Wu, Ao Tang, “An Even Tighter Bound for the Shapley-Folkman-Starr Theorem”, *Journal of Mathematical Economics*, V114, 2024. doi.org/10.1016/j.jmateco.2024.103028.

RESEARCH PROJECTS

- **Deep Reinforcement Learning for Integer-valued Action Space (Working)**
Keywords: Deep Reinforcement Learning, Autoregression, Quadratic Programming with Prof. Ozan Candogan
 - **Sequential Sampling:** We are developing a DRL approach to an integer-valued action space with a sequential action construction, rather than a one-shot decoder output.
- **Optimal Transport for a Multiclass $G/GI/N + GI$ Queue (Working)**
Keywords: Queueing Scheduling, Optimal Transport with Prof. Bahar Taskesen & Prof. Amy Ward
 - **Optimal Transport Approach:** We propose a new approach to the multiclass many-server queueing scheduling problem in the fluid limit, inspired by optimal transport theory. The reformulation inspired by optimal transport provides the optimal scheduling policy under a general discipline scheduling policy class in a finite horizon.
- **Dynamical Monotone System in Competitive Equilibrium Convergence (Working)**
Keywords: General Equilibrium, Supermodular Game with Prof. A. Kevin Tang
 - Building on the technical fluid-system results for dynamic monotone systems, we seek to provide an economic interpretation of the convergence conditions in a pure exchange economy.
- **Dynamic Competition in Elections: A Noncooperative Game of Candidate Positioning**
Keywords: Dynamic Game, Control Theory, Spatial Voting; (Submitted) with Prof. A. Kevin Tang
 - **Abstract:** A noncooperative dynamical model of elections is developed. We analyze the case of two candidates competing for a fixed, stationary electorate over a finite horizon. Equilibrium candidate trajectories are shown to be qualitatively determined by the interaction of three factors: voter distribution, campaign duration, and initial positioning.
- **An Even Tighter Bound for the Shapley-Folkman-Starr Theorem**
Journal of Mathematical Economics, V114, 2024 with Prof. A. Kevin Tang
 - **Abstract:** Based on a previous refined Shapley-Folkman lemma, we derive a tighter error bound for the Shapley-Folkman-Starr theorem and apply the result to the course allocation problem.
 - **Main Idea:** Instead of the traditional method of separately estimating each component of the Minkowski sum, we considered all components jointly and utilized the constraint on convexity from the refined Shapley-Folkman lemma to reach a tighter bound.

• **Multi-structured Signal Recovery Methods**

Jul. 2022 - Dec. 2022

Keywords: Signal Processing, Inverse Problem, Convex Opt.

Advisor: Prof. Jian-Feng Cai

- **Linear Combinations Approach:** Investigated recovery methods for multi-structured signals, such as sparse and low-rank matrices and tensors. Evaluated the optimality of linear combinations of recovery norms and assessed the effectiveness of the method of the square norm for low-rank tensors.
- **Atomic Norms Approach:** Identified the atomic norm as a recovery method for multi-structured signals. Explored the effectiveness of a new atomic norm and its corresponding atomic set, constructed from the intersection of known atomic sets for different structures.

• **Computational Optimal Transport**

Feb. 2022 - Jun. 2022

Keywords: Entropy, Sinkhorn Algorithm, Convex Opt.

Advisor: Prof. Jian-Feng Cai

- **Entropic Regularization:** Investigated the entropic regularization and extensively examined the local and global convergence properties of the Sinkhorn algorithm.
- **Sinkhorn Algorithm:** Explored the basic properties of the Sinkhorn algorithm. Implemented modifications to the Sinkhorn algorithm, such as the Greenhorn algorithm, with the aim of acceleration.

• **Shrinking a Convex Curve via Heat Equation**

Jun. 2021 - Aug. 2021

Keywords: Flow Geometry, Heat equation

Advisor: Prof. Tsz-Ho Fong

- **Rigorous Analysis:** Formal analysis of shrinking a convex curve to a point via the heat equation.

AWARDS

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| • Doctoral Centennial Fellowship | 2025 Fall |
| • Lee's Pharmaceutical - Kanya Lee Scholarship | 2023 Fall |
| • University's Scholarship Scheme for Continuing UG Students (Ranked 10%) & Dean's List | 2020-2024 |
| • Chern Class Talent Scholarship (For Math Students Pursuing Challenging Courses) | 2020-2024 |
| • Honorable Mention Award, Math Department, HKUST | 2023 |

TEACHING ACTIVITIES

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| • Grader of Honor Calculus I, HKUST | 2023 Fall |
| • Teaching Assistant at Math Support Center, HKUST | 2022 Fall |